



Artificial Intelligence Foundations (CIS4049-N)

Week 1 - Lab Introduction

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Lab Session - Introduction

- Introduction to R and R Studio

- R vs Python
- Familiarise with R and R Studio
- What is R and why using R – **Overview** and **Tutorial**
- Data Types – Vectors, Lists, Matrices, Arrays, Factors, Data Frames
- Variables, Operators, Decision-making, Loops, Strings, Functions
- Import Data in various formats – csv, Excel and Web Data
- Appendix – How to deal with errors in R



Lab Session - Introduction

- **This is not material that needs to be addressed and finalized during the 2-hour session!!**
- An almost complete tutorial on basic notions and structures of R, and you can use this material throughout the entire module and for the ICA as reference.
- If you decide to work with R in your ICA, it will be required to work on more advanced functions or packages (for example, on uninformed and informed search or reinforcement learning) and to define your own functions or R objects for the addressed problem.
- **Diagnostic Quiz**
 - https://forms.office.com/Pages/ResponsePage.aspx?id=WxHSQ16ltkad97AziOz8YCvNb6aBQQpAjKF_2wE9jYxUNI-E4STk4REJYOEW3TkVLWFIyNTNVNFZUMy4u
 - Not part of any assessments!!
 - It will help us understanding which topics we need to focus more in the postgraduate AI courses.
 - 25 Questions

Lab Session - Introduction

- If you have already worked with R, after completing and submitting the Diagnostic Quiz, since the main topic of the week is “**What’s AI**” – **an introduction to AI**, you can work on doing research on AI applications
- This will be part of the “Introduction” of your report (ICA), where you will introduce the addressed AI solution, and specify the main motivations behind your choice in the realm of the AI applications.

Questions?

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